

With Lily: Design and Implementation of Verbal and Nonverbal Interaction with a Virtual Humanoid in Mixed Reality for Affective Communication

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ABSTRACT

This demo presents a virtual humanoid that enables affective communication through verbal and nonverbal interaction in a mixed reality (MR) environment, showcased as an experiential exhibition titled *With Lily*. The aim of this work is to provide people—who desire social interaction yet feel burdened by concerns such as evaluation or misunderstanding—with a comfortable and enjoyable virtual conversation that is free from real-world social constraints. To this end, the demo implements a comprehensive system that integrates the technical design of verbal and nonverbal interaction for natural emotional and intentional exchange with an augmented virtual presence. The system further incorporates environment design, conversational narrative, and character personality design to foster users' communication motivation and emotional engagement.

Index Terms: Human computer interaction (HCI), Mixed / augmented reality, Intelligent agents, Interactive systems and tools

1 INTRODUCTION

People often seek emotional comfort through conversation. However, within real-world social relationships, concerns about evaluation or misunderstanding can discourage honest self-expression. Such circumstances highlight the need for a safe communication environment that allows for more open and uninhibited expression. In response, this demo proposes a virtual conversational partner embodied as a virtual humanoid that is independent of real-world social relationships.

The system focuses on embodied and affective expressions to support emotional communication and a sense of presence. Users' verbal and nonverbal inputs are interpreted by an LLM-based agent, and the resulting responses are mapped onto the virtual humanoid's facial expressions, behaviors, movements, and speech. In addition, we design a model-switching dialogue system that allows the humanoid's personality and conversational flow to change according to the context of the interaction. By further integrating environment design, narrative structure, and character personality design, the demo presents and implements a comprehensive design strategy for affective communication with a virtual entity.

2 EXHIBITION SETUP

2.1 Exhibition Space and User Experience

The demo space is configured as a room approximately 2 m × 3 m in size and includes objects such as a desk and a sofa that can be detected through the spatial awareness capabilities of the Meta Quest 3.

Participants wear an HMD and initiate a conversation while facing the virtual humanoid Lily, which is augmented into the physical space. During the experience, participants teach Lily, who is initially unfamiliar with the real world, nonverbal communication

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Figure 1: The participant wears an HMD to engage in the experience and initially faces the humanoid Lily seated on a white wooden stool. During the interaction, the participant can freely make use of the surrounding space.



Figure 2: The virtual humanoid Lily is responding to a participant's speech through speech and gestures while facing the participant and performing an open-handed presenting gesture.

methods, such as hand waving, as well as bodily actions, including walking and sitting. Participants also engage in virtual conversations by sharing personal stories. In response, Lily gradually adapts to the physical environment and continues the conversation by expressing herself more actively through facial expressions, gestures, and variations in vocal tone.

Through these interactions, participants experience emotional comfort afforded by conversations that do not persist in the real world, along with the enjoyment of expressing more candid emotions and forming a sense of intimacy with a virtual being.

2.2 Privacy and Consent

The demo temporarily stores audio and body movement data with the participant's consent for the purpose of maintaining interaction and conversational context. All collected data are deleted immediately after the experience and are not used for analysis or any form of reuse.

3 WHY A MIXED REALITY-BASED VIRTUAL HUMANOID

In this work, the role of the humanoid is not to perform functional tasks, but to engage in emotional communication and con-

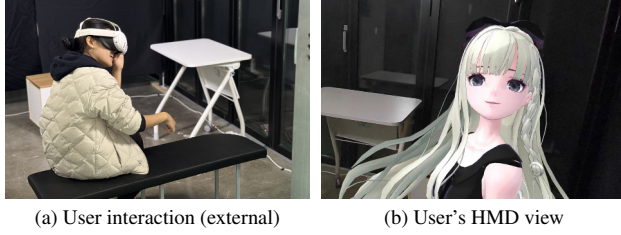


Figure 3: The interaction with Lily shown from two perspectives: (a) a participant interacting with the virtual humanoid Lily in the exhibition space, sitting side by side and speaking to her, and (b) the participant's HMD view, where Lily faces the participant and listens to the participant's speech.

nection with the user. Accordingly, the key requirement is that the humanoid actively responds to users' speech and actions and expresses emotions in a human-like manner. To achieve this, we adopt a virtual humanoid augmented within an XR environment, rather than a physical humanoid, which is constrained in its ability to move, express facial expressions and gestures.

Furthermore, by adopting mixed reality (MR), interactions take place within the user's personal space, reducing psychological burden. At the same time, the situation in which someone comes to the user creates a sense of intimacy, encouraging more active conversation.

4 INTERACTION MODALITIES

The system supports user inputs through speech, hand gestures, spatial and object-based pointing, and body-part-specific touch interaction. The virtual humanoid responds through speech, facial and behavioral expressions, spatial movement, object-based sitting and leaning actions, and contact-aware physical collision responses and body-part-specific touch recognition.

5 VERBAL AND NONVERBAL INTERACTION PIPELINE

Interaction with the virtual humanoid is primarily speech-based while simultaneously incorporating nonverbal expressions. During a user's utterance, nonverbal inputs—such as hand gestures, hand-based touch, and object and spatial pointing—are continuously collected. When the utterance ends, this moment serves as a trigger to integrate the accumulated nonverbal information collected during the utterance with the speech input transcribed via speech-to-text (STT). The integrated input is then passed to an LLM-based agent and interpreted according to predefined prompts.

The LLM agent subsequently generates a context-appropriate response in a predefined output format, which includes response,

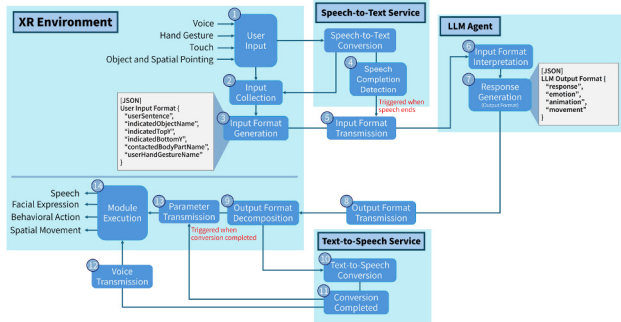


Figure 4: Overall Verbal and Nonverbal interaction pipeline from user input to module execution based on the LLM agent's output.

emotion, animation, and movement information. This output is then decomposed into individual components to control corresponding functional modules, resulting in spoken responses, facial expressions, behavioral actions, and spatial movements such as walking or sitting.

The demo operates in real time based on this interaction pipeline. In a single conversational turn, a latency of approximately 8–13 seconds occurs between the end of a user's utterance and the onset of the humanoid's response. This latency was observed in our current implementation using GPT-4.1 and ElevenLabs v3.

6 MODEL-SWITCHING DIALOGUE SYSTEM

In this research, we designed a model-switching dialogue system that dynamically switches between multiple LLM models with different predefined prompts according to the conversational context. The system consists of a stage-based model and situation-specific models.

In our demo, we endow the virtual humanoid with a timid personality in the initial stages and gradually transition it to a more proactive personality in later stages, creating an experience of progressively developing intimacy. In addition, under specific conditions, the humanoid guides conversations toward real-world topics or intentionally steers the dialogue into predefined scenarios—such as asking how a real-world hand greeting is performed—thereby enabling more diverse communication.

7 CONCLUSION

Based on observations from the With Lily demo, users tended to participate more actively in conversations after recognizing that their nonverbal expressions were reflected in the interaction. This suggests that nonverbal expression can function not only as support for verbal communication, but also as a meaningful factor enhancing conversational engagement and emotional immersion.

Meanwhile, although many users were not fully aware of the extent to which their nonverbal expressions were reflected in the system, this uncertainty encouraged a wider range of expressions, contributing to more diverse interactions. In addition, the narrative context of the virtual humanoid—framed as a being encountering the real world for the first time—helped users interpret occasional unnatural responses caused by technical limitations as aspects of the character's personality, thereby sustaining immersion.

Based on these observations, this demo proposes a virtual humanoid in mixed reality as an approach to affective communication and emotional connection with a virtual entity, while presenting the technical and experiential design strategies explored in the demo.

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